Global Facilities - Design & Construction Standard - Mechanical - Process Vacuum System

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1.0 Purpose

1.1 This document serves as a System Standard document that provides specifications that shall be followed during the Design, Engineering, and Construction of new Process Vacuum System as part of new Greenfield FAB Construction projects.

1.2 The standard is applicable for Process Vacuum system.

1.3 The document is intended to supplement the actual Construction Drawings and Construction Specifications for Construction projects.

2.0 Scope

2.1 This document covers the new Basebuild System installations only, as part of new Greenfield FAB Construction projects. This document does NOT cover Tool Installation or Point-of-Use (POU) Abatement.

2.2 This document does not cover other Vacuum systems, such as Arsenic Vacuum system.

2.3 This document applies to all Micron Facilities employees, contractors, and vendors at all Micron sites.

3.0 Definitions and Acronyms

Acronym	Definition / Description
AE	Acid Exhaust
AHJ	Authority having Jurisdiction
AT	Assembly and Testing
CHW	Chilled Water
FCT	Facilities Central Team. This is used interchangeably with Global Facilities Engineering Team and associated to O&M BKMs
FE	Front-EndEnd
FMCS/SCADA	Facility Monitoring and Control System / Supervisory Control and Data Acquisition
GFC	Global Facilities & Construction Team. This is used interchangeably with Global Facilities (GFac), Facilities Central Group (FCG), World Wide Operations (WWOps), World Wide Facilities (WW_FACILITIES), Corp Facilities
GFTT	Global Facilities Technology Team. This is used interchangeably with Global Facilities Engineering Team and associated to design standards
GHVAC	General Heating, Ventilation, Air Conditioning
Global Facilities Central Team	This is used interchangeably with Innovation & Integration Team
Global Facilities Construction Team	This is used interchangeably with Global Construction Team
Global Facilities Operations Support Team	This is used interchangeably with Global Facilities Operations Team
Global Facilities Planning Team	This is used interchangeably with Global Facilities Line Planning Team
HAP	Hazardous Air Pollutants
LCP	Local Control Panel
LOTO/COHE	Lockout Tagout / Control of Hazardous Energy
N+1+F	System sizing concept for equipment quantity consisting of N duty units, +1 redundant standby unit, + future unit(s)
PLC	Programmable Logic Controller
POC/LPOC	Point of Connection / Lateral Point of Connection
POU	Point of Use
PV	Process Vacuum
TUM	Tool Utility Matrix
UPS	Uninterruptible Power Supply
VSD/VFD	Variable Speed Drive / Variable Frequency Drive

4.0 Roles and Responsibilities

Roles	Responsibilities
GFTT Lead (Discipline Engineer)	• Develop and maintain System Standard document • Ensure Best Known Methods (BKM), lessons learned, and emerging technologies are documented in System Standard • Perform periodic document review and update System Standard
FCT Lead (Discipline Engineer)	• Review and provide technical feedback on System Standard • Support the Site teams by providing knowledge and technical expertise
Site Project Construction Teams	• Review and provide technical feedback on System Standard • Incorporate System Standard into Project Engineering, Design, and Construction • Document project deviations, exceptions, and exclusions from System Standard • Share new lessons learned to GFTT lead, to determine if it is Best Known Method (BKM) that can be captured in future revision of System Standard
Site Facilities Teams	• Use System Standard as technical reference

5.0 Overview of Systems

The following Process Vacuum systems are used to support the requirements from Manufacturing and Facilities loads:

5.1 **Process Vacuum (PV)** is vacuum used to hold wafers on to tool chucks.

The Process Vacuum System is designed based on the concept of providing sufficient amount of process vacuum air at specified vacuum levels to meet the demands from the various production tools.

The main Process Vacuum System equipment shall be as follows:

5.1.1 Process Vacuum Pump

5.1.2 Vacuum Receiver Tank

6.0 Performance Specs

6.1 The Process Vacuum Systems shall be designed and installed to achieve the operational performance specs as outlined below; See **Table 1**.

Table 1: Operational Performance Specs for Process Vacuum Systems

Parameter	Specification Requirement	Remarks
Process Vacuum Pressure	< 133 mbar (abs) (< -26 in Hg Gauge or < -88 kPa Gauge)	Measured at header on PV equipment or Vacuum Receiver Tank.

Note: The Process Vacuum Systems shall be designed and installed with capability to meet the Process Vacuum pressures stated above. After installation, the actual operation of the systems and setpoints for system pressures at receiver tank may be different (less) than the pressures shown. This is due to different loading of Fab at different stage of Fab.

7.0 Equipment Sizing & Design Requirements

7.1 **Redundancy & Future** – Process Vacuum System main equipment shall be designed for N+1+F for Process Vacuum Pumps. All main system equipment shall have N+1 redundancy (with one redundant unit having capability as online standby), and a provision to install additional future units (F) for future capacity expansion.**System Diversity and Safety Factor –** Process Vacuum System sizing shall be selected so that the N capacity meets 115-120% of the process vacuum demand at full build (after initial construction of the system), based on the manufacturing tool layout and TUM, plus the Facilities Base Build requirements. Therefore, the Process Vacuum system sizing capacity for capacity shall have 15-20% spare future capacity over the process vacuum demand at full build (after initial construction of the system).

7.2.1 To determine the Process Vacuum demand at N full build, a diversity factor of 80% of the Peak demand from the TUM shall be used. In addition, the Process Vacuum system shall be sized to include the capacity of future units (F).

See **Table 2** for sizing criteria and calculation.

Table 2: System Sizing Calculation – Process Vacuum System

- Process Vacuum System Demand at Full Build, D:
  D = (TUM Peak Demand x 80% Diversity) + Facilities Basebuild Requirements*
- Process Vacuum System N capacity:
  N = 115-120% x D
- Process Vacuum System Sizing to include future units:
  Sizing = N + F

7.3 **Equipment Quantity –** Process Vacuum System main equipment shall be selected for a minimum of 3 units (2+1) or more, with each unit to be equal in capacity sizing; for process vacuum pumps. The process vacuum demand at full build shall be split evenly across the number of equally sized process vacuum equipment.

7.4 **Equipment Arrangement** – All Main equipment shall be arranged in parallel together and connected via common piping headers for each equipment. This allows any combination of equipment to be in operation concurrently and allow for redundancy on all equipment.

7.5 **Equipment Layout –** Equipment shall be suitably located to comply with all equipment maintenance clearances and access requirements. Adequate space and a clear routing path shall be reserved for future move-in of new equipment, and to allow move-in and move-out of large components during maintenance and repair (such as PV pumps).

7.6 **Equipment Selection Criteria** – See **Tables 3** for Process Vacuum system selection criteria and requirements.

Table 3: Equipment Selection Criteria

<table>
<tr><th>Equipment</th><th>Type & Description</th></tr>
<tr><td>Process Vacuum Pump</td><td>

Pump type shall be either oil lubricated rotary vane, liquid ring type, or dry rotary screw, all type of PV pump shall be direct driven.
Characteristic of different PV pump are shown in below:

<u>Oil Lubricated Rotary Vane Pumps:</u>

Typical Operating Range: > 1.33 mbar(abs) or -29 in Hg
Ability to handle process carryover: Low
Utilities required: None
Initial Capital Cost: Low
Repair Difficulty: Low

<u>Liquid Ring Water Sealed Pumps:</u>

Typical Operating Range: > 40 mbar(abs) or -28 in Hg
Ability to handle process carryover: High
Utilities required: seal cooling water
Initial Capital Cost: Medium
Repair Difficulty: Low

<u>Dry Rotary Screw Pumps:</u>

Typical Operating Range: > 0.133 mbar(abs) or -29 inHg
Ability to handle process carryover: Low to Medium
Utilities required: None (subject to manufacturer requirement)
Initial Capital Cost: High
Repair Difficulty: High

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	• Cooling method shall be water-cooled type (all wet type vacuum pump) • Process Vacuum (PV) Pump sizing and selection shall be as follows: ○ Pumps shall be selected with the performance indicated at the design operating point (flow and pressure, at partial VSD speed and motor brake-horsepower). ○ Pump curve shall show the performance indicated at the maximum operating point (flow and pressure, based on pump screw/vane size, VSD speed, and full brake-horsepower to match motor size rating). ○ Sound level performance to be < 80 dBA at distance of 1m away from pump enclosure. ○ Pumps shall be selected based on actual flow conditions. Formula below is conversion of standard flow to actual flow for ambient air in SI unit. $$\underset{\text{(ACMH)}}{Q_{actual}} = \underset{\text{(SCMH)}}{Q_{std}} \times \frac{P_{std} \times T_{actual}}{P_{actual} \times T_{std}}$$ ○ Pumps shall be selected based on adjusted vacuum rating due to elevation impact. $$\text{Adjusted Vacuum Rating} = \text{Actual Atmospheric Pressure} \times \frac{\text{Nominal Vacuum Rating}}{\text{Standard Atmospheric Pressure}}$$ **Note:** Performance of a vacuum pump is mainly affected by the altitude due to atmospheric pressure vary at different elevation.

- PV Pump motor sizing and selection shall be as follows:
  - Pump motors shall be selected for non-overloading operation on all points of the performance curve at the design conditions.
  - Pump motors shall be rated for inverter duty with variable speed operation.
  - Pump motor efficiency rating shall be minimum IE3.
  - Pump motor IP rating shall be minimum IP55.

- PV Pump installation shall be equipped with following:
  - Inertia base and vibration isolation springs.
  - Oil cooler*
  - Oil separator*
  - Lube oil system for bearing/gearbox, etc.
  - Cooling system for pump system
  - Seals to prevent contamination from oil/vapour.
  - Inlet check valve.
  - Exhaust filter.

* Applicable to oil lubricated vacuum pump only.

Note: Approved Manufacturers – Busch, Pfeiffer, Elmo Rietschle, Gardner Denver Nash, Sihi, Atlas Copco or approved equivalent

<table>
<tr><td>Process Vacuum Receiver Tank</td><td>

Receiver Tank shall be located before the PV pumps.
Receiver Tank shall be vertical mounted type, with standalone legs.
Receiver Tank sizing and selection shall as follows:
Receiver Tank capacity shall be $0.3m^3$ volume per 1,000 CMH of Total PV System Capacity (N+F, CMH) or minimum $4m^3$ volume.
$$\text{Receiver Tank Capacity}\,(m^3) = \frac{\text{PV System Capacity}}{(\text{N+F, CMH})} \times \frac{0.3\ m^3}{1{,}000\ \text{CMH}}$$

Receiver Tank shall be stainless steel or Epoxy coated carbon steel.
Receiver Tank shall be rated for pressure vessel ASME Section VIII Division 1.
Manhole shall be blind flange of minimum neck manway access size of 16 inch or more with tight gasket sealing and bolts.
The building structure and floor load rating shall be capable of allowing on-site hydrostatic pressure testing of the Receiver Tank to be performed (via water fill of the tanks) - for minimum 1 tank at a time.
Receiver Tank to be pneumatic tested 1.3 times of the positive pressure or hydrostatic tested 1.5 times of the positive pressure.

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<tr><td>Inlet Filter</td><td>

Inlet Filter shall be installed before the PV pumps.
Inlet Filter can be integrated within the PV pumps as well.
Filter shall be minimum of 99.7% filtration efficiency at 10 microns.

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8.0 Piping Design and Sizing Requirements

8.1 **Piping Diversity and Sizing Factor** – The Process Vacuum system main piping shall be sized according to the same sizing factors (N+F, System Diversity, etc.) stated in the above section.

8.2 Each PV lateral shall be sized so that the lateral's capacity meets 115-120% of the PV demand at full build for the lateral (after initial construction of the system), based on the manufacturing tool layout and TUM. Therefore, each PV lateral shall have 15-20% spare future capacity over the PV demand at full build for the lateral (after initial construction of the system). **Table 4.**

Table 4: PV Pipe Sizing Calculation

- PV Main Piping Sizing = N + F
- PV Lateral Demand at Full Build,
  D_L = TUM Peak Demand for Lateral x 70-80% Diversity
- PV Lateral Piping Sizing = 115-120% x D_L

8.3 **Pipe Sizing Criteria –** PV Pipe Sizing shall be selected based on the full build capacity of the PV system equipment, including the future equipment (N+F). The Pipe Sizing shall be designed with maximum flowrate not to exceed the tables below: See **Table 5**.

Table 5 – Pipe Sizing Criteria for PV

PV Nominal Pipe Size	Maximum Flowrate			Designed Velocities (m/s)	Approximate Pressure Loss
	SCFM	NCMH	SLPM		
1″ DN25	6	23	167	6	6.7 Pa/m
2″ DN50	48	169	1,372	12	7.9 Pa/m
2.5″ DN65	70	281	1,969	12	5.9 Pa/m
3″ DN75	108	523	3,068	12	4.1 Pa/m
4″ DN100	283	1,123	8,011	18	6.1 Pa/m
6″ DN150	641	3,417	18,163	18	3.4 Pa/m
8″ DN200	1,124	7,236	31,817	18	2.3 Pa/m
10″ DN250	1,768	13,185	50,056	18	1.7 Pa/m
12″ DN300	2,501	20,803	70,821	18	1.4 Pa/m
14″ DN350	4,961	26,120	140,467	30	3.5 Pa/m
16″ DN400	6,508	37,208	184,273	30	2.9 Pa/m
18″ DN450	8,260	50,709	233,900	30	2.6 Pa/m
20″ DN500	12,332	85,114	349,201	30	2.0 Pa/m
22″ DN550	14,982	109,350	424,235	30	1.8 Pa/m
24″ DN600	17,770	136,154	503,186	30	1.6 Pa/m

8.3.1 If the pipe sizing is designed with velocities exceeding Table 5, the piping system pressure drop calculations shall demonstrate that the PV pump selections have adequate performance rating to meet the design parameters and system requirements.

8.4 Materials of Construction

8.4.1 **Process Vacuum Pipework –** PV Distribution piping shall be UPVC schedule 80.

- Type 16 – Pipe standard to ASTM D1785.
- Fittings shall refer to ASTM D2467 schedule 80 for socket fittings.
- Fittings shall be solvent welded using solvent cement recommended by the fitting manufacturer.
- PV piping shall be fully welded as much as possible to minimize the number of joints and flanges.

8.4.2 **Chilled Water (CHW) Pipework –** CHW piping shall be A53 Carbon Steel or ASTM A53B SCH 40 galvanized steel.

- CHW piping shall be fully wedded as much as possible, to minimize the number of joints and flanges.

8.4.3 **Process Vacuum Exhaust Ductwork** – Ductwork shall be FM Approved Fluoropolymer-coated Stainless Steel, FM Approved Fiberglass Reinforced Plastic (FRP), or standard Fiberglass reinforced Plastic (requiring internal sprinklers) materials of construction or approved equivalent.

- Fluoropolymer-coated Stainless-Steel ductwork includes ETFE (Ethylene-Tetrafluoroethylene) and ECTFE (Ethylene-Chlorotrifluoroethylene) coating materials.
- It is recommended to use FM Approved ductwork materials, in lieu of standard FRP ductwork that is not approved by FM. Ductwork that is FM Approved per FM Standards 4922 and 4910 do not require internal sprinklers, per NFPA Standard 318 and FM requirements.
- **Note:** Ductwork made of PVC, PP or PE shall NOT be allowed for sizes 6” and larger. The exception is smaller size PVC ductwork, sizes 4” (DN100) and below, which is typically used for tool hookup ductwork and VMB ductwork.
- **Gaskets, general:** In general, PTFE (Polytetrafluoroethylene, aka Teflon) tape gaskets or rope gaskets shall be used for Acid Exhaust. Gasket shall be installed on the duct flange lip surface.
- **Gaskets, special cases:** For special areas in which there is potential for liquid accumulation or condensation, it is recommended to use Viton Full-Face type gaskets, 1/8” (3.0mm) thick. The Full-Face type gasket shall be installed on the entire flange ring face of the angle rings, at each ductwork joint.

8.4.4 **Process Vacuum Exhaust Assignment** – Due to potential process carryover, PV Exhaust shall assign to compatible Process Exhaust System.

- For FE site, PV Exhaust Duct shall assign to Acid Exhaust System due to potential process carryover and potentially corrosive.
- For AT site, PV Exhaust Duct recommended to assign to Scrubber Exhaust System due to potential process carryover. However, due to site facilities constraint, PV Exhaust Duct allow to assign to General Exhaust System. (Subject to Global Facilities Team review and approval.)

8.5 Spare Valves for Main Expansion

8.5.1 Full-size valves shall be installed at all end-of-line at the main plantroom headers for PV pumps, to allow for future expansion.

8.6 Spare Valves for Future Laterals

8.6.1 Additional isolation valves shall be included on the main supply distribution piping, to allow installation of future laterals for capacity expansion. Based on the number of laterals included in the design, a minimum quantity of 35-50% of additional isolation valves for future lateral shall be included.

8.7 Lateral Point-of-Connection Quantity and Sizes

8.7.1 The laterals shall have sufficient quantity of Point of Connections (POCs) to meet the tool layout and requirements.

8.7.2 A minimum quantity of 30-40% of additional spare POCs shall be provided on each lateral, for future tools.

8.7.3 The sizes for each POC shall be checked and verified versus the actual tool requirements. Minimum POC size shall be DN25 (1”), but larger POC sizes can be provided, based on the actual tool requirements.

9.0 Sustainability Design

9.1 **VFD for PV Pumps -** All pumps shall be VFD driven for variable speed operation to optimize energy consumption in order to run PV pump at partial loading.

9.1.1 In case of loss of control signal, VFD shall be Fail-Last Speed Configuration.

9.1.2 VFD Recommended SCADA Parameters and Alarm Set Points shall refer to “Global Facilities – Design & Construction Standard - Electrical – Electrical Design Specification”.

9.2 **High Efficiency Motor** – PV Pumps motors shall be National Electrical Manufacturers Association (NEMA) or IE3 above to reduce energy consumption and CO2 emission.

10.0 Accessories

10.1 **Other Accessories -** Provide other accessories as outlined in **Table 6**.

Table 6: Accessories to be installed

Accessories	Type & Installation Location	Remarks
Piping Insulation	• Install for all CHW piping throughout the system	In accordance with Spec Section 15260 (or equivalent Spec)
Piping Support	• Install piping hangers, supports, racks, and clamps at the appropriate spacing interval as required by the system and piping load and layout • Provide seismic restraints and supports as deemed required during design, based on site location and local jurisdiction requirements • Provide vibration isolation for piping supports as deemed required during design, based on the building vibration criteria	In accordance with Spec Section 15140 (or equivalent Spec)
Ductwork Supports	• Install duct hanger and support material gauges in compliance with details on the Drawings. Where details are not provided, comply with SMACNA Industrial Duct Construction Standards. • Ductwork hanger and suspension material may be threaded rod, structural shapes, or strut material. The use of wire as suspension material is prohibited. • Ductwork hanger supports and spacing shall take into account the potential weight of water when the ductwork contains fire sprinklers.	In accordance with Spec Section 15140 (or equivalent Spec)
Flexible Connections	• Install at all PV connections to PV Pump on suction. • Install at all Exhaust connections to PV Pump on discharge. • Install at all CHW connection to PV Pump Oil Cooler Heat Exchanger on supply and return.	
Pressure Gauges	• Install at each PV Pump suction. • Install at each Receiver Tank • Install Differential Pressure gauge on each filter • Install at CHW supply and return piping to Oil Cooler Heat Exchanger.	
Pressure Transmitter	• Install at Main Header end-of-line before Receiver Tank. • Install at Receiver Tank. (Quantity: 2 total, 1 duty, 1 redundant) • Install at Lateral end-of-line on PV for a minimum of 2 laterals (2 longest distance lateral)	

Temperature Gauges	• Install at each PV Pump Oil Cooler Heat Exchanger inlet and outlet on the CHW side.	
Air vents	• Air vents shall be minimum size DN20 (3/4 inches) • Install Auto or Manual Air Vents at all high point locations for CHW.	
Drain Ports	• Install at all Receiver Tanks.	
Valves	**<u>PV</u>** • Valves shall be PVC body, EPDM class rubber seal, full bore ball valve with handle parallel to the direction of flow when the valve is fully open. • Install balancing valve or manual chilled water flow control valve at Oil Cooler Heat Exchanger outlet. **<u>CHW</u>** • Valves DN80 (3 inches) and larger shall be ductile iron body, Ethylene Propylene Diene M (EPDM) class rubber liner, stainless steel disc and stem, lug style butterfly valves with position locking handle. • Wafer type butterfly valves shall NOT be used in CHW and CW systems. • Instrumentation points of connection shall be provided with a means of isolation (ball valve, etc.) for maintenance and/or replacement.	In accordance with Spec Section 15101 (or equivalent Spec)
Control Valves	• Provide all control valves with manual override function. • The PV Control valve actuators and positioners shall provide position feedback status of the actual valve position.	
Labeling and Identification	• Provide identification labels at minimum of every 6m distance and change of direction **<u>Piping</u>** • Labels on PV Piping shall be Blue Background with White Lettering • Labels on CHW Piping shall be Green Background with White Lettering • “PV” nomenclature shall be used for Process Vacuum • “CHWS” and “CHWR” nomenclature shall be used for Chilled Water Supply and Return **<u>Ductwork</u>** • Labels on Exhaust Ductwork shall be Yellow Background with Black Lettering • “AE” nomenclature shall be used for Acid Exhaust	In accordance with Spec Section 15190 (or equivalent Spec)

11.0 Electrical Design

11.1 **Power Segregation –** Electrical power supply for PV Pumps shall be segregated among a minimum of 2 separate upstream power sources (including panel + transformer). The quantity of equipment served from each upstream power source shall be split evenly.

11.2 **Un-Interrupted Power for Pumps -** 50% of PV pumps shall be connected to an Un-interrupted Power source (such as UPS or DC Bank) to allow continuous operation and ride-through capability of the PV pumps during voltage sags, spikes, and fluctuation events.

11.3 **Emergency Power for Pumps -** PV pumps not required emergency power source.

11.4 **Un-interrupted Power for Controls –** 100% of PV system controls including PLCs, Local Control Panels (LCPs), and other auxiliary control power shall be connected to an Un-interrupted Power (UPS) source to allow continuous operation of the Process Vacuum system during momentary power interruption or voltage sags, spikes, and fluctuation events.

11.5 **Variable Frequency Drives –** All pumps shall be VFD driven for variable speed operation. Variable Frequency Drives shall be selected to be one size larger than the power rating of the actual pump motor size.

11.5.1 Variable Frequency Drives shall be compliant with SEMI standard F47 for voltage sag immunity, to provide ride-through capability during momentary power interruption or voltage sag.

11.6 **Motor Efficiency –** PV Pumps motors shall be National Electrical Manufacturers Association (NEMA) or IE3 above.

11.7 **Local Disconnect Switches –** Each PV Pumps shall have a lockable type, local disconnect switch installed adjacent to the pump, to allow for manual isolation and lockout/tagout (LOTO/COHE) of the pump.

12.0 Instrumentation

12.1 The following Instrumentation shall be installed and monitored/controlled via FMCS: See **Table 7**.

Table 7 – Instrumentation Devices

Accessories	Type & Installation Location
Pressure Transmitters	• Install at Vacuum Receiver Tank (Quantity: 1 duty and 1 backup [2 Total]) • Install at Main Supply Header after Vacuum Receiver Tank (Quantity: 1 duty and 1 backup [2 Total]) • Install at PV lateral end-of-line, at minimum 2 locations at furthest end of fab to allow online monitoring. (Quantity: minimum 2 locations)
Exhaust Flowmeters	• Install a flowmeter on exhaust sub main within PV Plant to allow continuous exhaust load monitoring.
Additional Spare Ports	• All PV laterals shall have a spare port and installed at the lateral end-of-line, to allow for future pressure transmitters to be installed. The future pressure transmitters will allow continuous SCADA monitoring of each lateral pressure.

12.2 Instrumentation points of connection shall be provided with a means of isolation (ball valve, etc.) for maintenance and/or replacement.

13.0 Control Sequence Philosophy

13.1 The overall philosophy for system control and operation shall be listed in this section. See **Table 8**.

Table 8: System Control Logic

Control Parameter	Control Logic
Pump Pressure Control	• Speed output of Pump VSDs shall be modulated to maintain the pressure setpoint, as measured by main pressure transmitters, located at Process Vacuum Receiver Tank.

13.2 A detailed Sequence of Operation (SOO) shall be developed specifically for each system, as part of each construction project.

13.3 Control Valves

13.3.1 The control valve fail positions shall be Fail-Last configured.

13.3.2 The control valves shall be double-acting type actuators, capable of failing to last state position and holding the position for unlimited duration without needing control air, power, or PLC signal.

13.3.3 After a loss event of control air, power, or PLC signal, and after it is restored, the control valve output command/position shall be set to the same command/position it was operating at prior to the loss event.

13.3.4 All control valves shall have a manual override bypass, to allow manual operation of the damper upon a failure.

14.0 Smart Facilities Design

The following are additional enhancements that can be considered and included to enable Smart Facilities Design concepts.

14.1 **Lateral Pressure Online Monitoring –** Consider installing pressure transmitters at all PV laterals at each end-of-line location, to allow continuous SCADA monitoring and alarm of each lateral pressure.

14.2 **Load Demand Online Monitoring –** Consider installing flowmeters at the main headers or at each major load/user, to allow continuous SCADA monitoring and tracking of PV load.

14.3 **Pump Vibration Online Monitoring –** Consider installing vibration sensors on each PV pump and motor, to allow continuous monitoring and alarm of pump vibration levels.

15.0 Inter-Discipline Coordination Matrix

15.1 The following provisions shall be provided by each of the following disciplines and coordinated accordingly: See **Table 9**.

Table 9: Inter-Discipline Coordination Matrix

Discipline	Provision
Architectural	• Provide dedicated Utility Room or Mechanical Room space for centralized Process Vacuum equipment • Provide concrete housekeeping pad below all equipment including Pumps, Receiver Tank, etc. • Cut openings in walls and floors for ducting penetrations, and seal openings after piping or ducting installation • Provide adequate fire stopping at all required fire wall penetrations
Mechanical	• Provide HVAC space conditioning for dedicated Utility Room or Mechanical Room space containing the centralized Process Vacuum equipment • Provide chilled water piping and accessories as outlined in this Standard • Provide exhaust ductwork and accessories as outlined in this Standard • Provide drainage and accessories as outlined in this Standard
Electrical	• Provide VFDs for all pumps • Provide local disconnect switches for all pumps • Provide power wiring for all pumps and Electrical equipment/ components • Provide grounding for all pumps and Electrical equipment/ components • Provide Emergency Power / UPS Power as outlined in this Standard • Provide Power Segregation as outlined in this Standard
Controls	• Provide control valve actuators and positioners • Provide control wiring • Provide instrument control air tubing • Provide FMCS/SCADA system monitoring and controls of all instrumentation and transmitters including pressure, temperature, flow, etc. • Provide FMCS/SCADA system connectivity of all Local Control Panels or Master Control Panel

16.0 Quality Inspection and Testing

16.1 **Inspection and Test Plan (ITP) –** Create formal Inspection and Test Plans (ITP) that outline all requirements for quality verification, inspections, testing, and commissioning:

16.1.1 Piping and Ductwork

- Piping and Ductwork Shop Drawings and Submittals
- Incoming Material Inspection
- Material Storage
- Welding Procedures, Qualification and Inspection
- Assembly Procedures, Qualifications, and Inspection
- Ductwork coating spark testing
- Piping and Duct Supports Installation and Verification
- QA/QC Inspection & Punchlist
- Ductwork Leak Testing – A minimum of 25% of Exhaust fume ductwork shall be leak tested.
- Vacuum Leak Testing – system operating pressure (-26 in Hg) for 4 hours.
- Quality Certification
- Final QA/QC Inspection & Punchlist
- Training, Documentation, and Handover

16.1.2 Equipment

- Equipment Shop Drawings and Submittals
- Factory Acceptance Test
- Incoming Equipment Inspection
- Equipment Move-In and Installation
- QA/QC Inspection & Punchlist
- Equipment Pre-Startup Checks – Mechanical, Electrical, Controls, etc.
- Operational Acceptance Test (OAT) - Standalone Equipment Startup per Vendor's Procedure to allow equipment to run in MANUAL mode. Also considered "Early Run" Equipment.
- Final QA/QC Inspection & Punchlist
- Training, Documentation, and Handover

16.1.3 System

- Site Acceptance Test (SAT) – Test System operation and control functions in accordance with P&ID, SOO, and Specifications.
- Integrated System Test (IST) – Test full System operation including all system interdependencies, interlocks, and critical failure modes.
- Final QA/QC Inspection & Punchlist
- Training, Documentation, and Handover

16.2 **QA/QC Inspection** – Prior to vacuum leak testing, QA/QC inspection shall be performed on all installed piping and components to verify proper installation in accordance with construction drawings, project specifications, and industry standards.

16.3 **Vacuum Leak Test** – Vacuum Leak Test shall perform at -26 in Hg for 4 hours. Vacuum leak test shall be performed according to “Global Facilities – General Plastic Piping Pressure Testing Standard”. Walk the entire system to verify that no leaks are present, and to make repairs to any leaks that are identified.

17.0 Startup Commissioning / Site Acceptance Test

17.1 At a minimum, the following system control functions shall be tested and successfully proven during the System Startup Testing & Commissioning / Acceptance Tests.

17.1.1 Pumps

- Pump Motor Rotation Check
- Pump/Motor Alignment Check and Lubrication
- VSD Settings and Functionality Checks
- Control interface and communication checks to FMCS/SCADA
- Pump Standalone Startup Procedure per Manufacturer’s procedure
- Pump Vibration Level Checks
- Verify Pump Operation through its operating range (0 to 50/60 Hz)
- Verify Pump Start/Stop Function to Rotate Operating Pump
- Verify Pump Auto-Start sequencing, upon failure of an operating Pump
- Verify Pump Speed control in response to Vacuum pressure fluctuations
- Verify Pump Inlet Motorized Damper opens upon pump startup and closes upon pump stop, while keeping vacuum pressure stable
- Verify Pump oil temperature per Manufacturer’s recommendation by setting manual flow control valve to supply the required CHW flow to Pump Oil Cooler.
- Confirm Process Vacuum system pressure control is stable during pump manual start/stop rotation
- Confirm Process Vacuum system pressure control is stable during pump fault/trip event and next pump auto-start sequencing
- Confirm Process Vacuum Rotation Function, Auto Sequencing, and stable system pressure control via the Master Sequencer Central Control Panel.
- Verify control interface and communication checks to FMCS/SCADA

17.1.2 Instrumentation and Controls

- Confirm all control valves fail to the stated position per this Standard, upon loss of instrument air, power, and/or control signal.
- Verify all alarms are enabled and activated in SCADA.

18.0 Reference Specifications

18.1 The following Construction Specifications shall be referenced during design, engineering, and construction. The Specifications shown below are in accordance with the Construction Specifications Institute (CSI) Master Format; Refer to the specific Construction Specifications for each project. See **Table 10**.

Table 10 – Specification Sections

Spec Section	Spec Contents
Section 40 05 06	Couplings, Adapters, and Specials for Process Piping
Section 40 05 07	Hangers and Supports for Process Piping
Section 40 05 31	Thermoplastic Process Pipe
Section 40 05 60	Valves
Section 43 11 43	Gas Handling Vacuum Pumps
Section 40 70 00	Instrumentation for Process Piping
Section 40 05 97	Identification for Process Equipment
Section 40 05 96	Vibration and Seismic Controls for Process Equipment
Section 23 05 00	Common Work Results for HVAC
Section 23 07 19	HVAC Piping Insulation
Section 23 21 13	Hydronic Piping
Section 23 21 16	Hydronic Specialties
Section 23 09 00	Instrumentation for HVAC Piping
Section 23 31 13	Metal Ducts
Section 23 33 00	Air Duct Accessories
Section 23 05 93	Testing, Adjusting, and Balancing for HVAC
Section 40 05 93	Common Motor Requirement for Process Equipment
Section 40 97 00	Variable Frequency Drives

19.0 Document Control

19.1 **Approval**: GLOBAL_FAC_GFTT_APPR

19.2 **Notification**

GLOBAL_FAC_MANAGERS
GLOBAL_FAC_COMMUNICATIONS
GLOBAL_FAC_CONSTRUCTION
GLOBAL_FAC_NOTIFY
FCG_F2_FAC_NOTIFY
FCG_F4_FAC_NOTIFY
FCG_F6_FAC_NOTIFY
FCG_F10_FAC_NOTIFY
FCG_F11_FAC_NOTIFY
FCG_F15_FAC_NOTIFY
FCG_F16_FAC_NOTIFY
FCG_MMP_FAC_NOTIFY
FCG_MMY_FAC_NOTIFY
FCG_MSB_FAC_NOTIFY
FCG_MTB_FAC_NOTIFY
FCG_MXA_FAC_NOTIFY

19.3 **Final Viewer Access**

All Micron Team Members at all Sites

19.4 **Retention**

Adherence to the requirements specified in
Global Facilities - Document Control Procedures

19.5 **Review**

Biennially (two years)

20.0 Revision History

Revision #	Section Changed	Details of Changes	Date
0	NA	New document	03/11/21